Practical - 3

Mathematical Equations

1 LaTeX Code

The following is the complete LaTeX source code used to create this practical document.

```
\documentclass[12pt,a4paper]{article}

% Packages
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{lmodern}
\usepackage[a4paper,margin=1in]{geometry}
\usepackage{amsmath,amssymb}
\usepackage{xcolor}
\usepackage{graphicx}
\usepackage{listings}
\usepackage{fancyhdr}

% Code colours and style
\definecolor{codegreen}{rgb}{0,0.5,0}
\definecolor{codegray}{rgb}{0.5,0.5,0.5}
\definecolor{backcolour}{rgb}{0.95,0.95,0.95}

\lstdefinestyle{latexstyle}{
    backgroundcolor=\color{backcolour},
    commentstyle=\color{codegreen},
    keywordstyle=\color{blue},
    numberstyle=\tiny\color{codegray},
    basicstyle=\ttfamily\scriptsize,
    breaklines=true,
    keepspaces=true,
    numbers=left,
    numbersep=8pt,
    showstringspaces=false,
    frame=single,
    tabsize=4
}

\lstset{style=latexstyle}

% Header and footer
\pagestyle{fancy}
\fancyhf{}
\fancyhead[L]{Mathematical Equations}
\fancyhead[R]{Practical - 3}
\fancyfoot[R]{Page \thepage}
\renewcommand{\headrulewidth}{0.4pt}
\renewcommand{\footrulewidth}{0.4pt}

\setlength{\parindent}{0pt}
\setlength{\parskip}{6pt}

\begin{document}

% Title Page
\begin{titlepage}
    \centering
    \vspace*{3cm}

    {\Huge\bfseries Practical - 3\par}
    \vspace{1cm}

    {\LARGE Mathematical Equations\par}
    \vspace{1cm}
    \vfill
\end{titlepage}

% LaTeX Source Code
\section{LaTeX Code}

The following is the complete LaTeX source code used
to create this practical document.
```

```
69
70  \lstinputlisting[
71      style=latexstyle,
72      language={[LaTeX]TeX}
73  ]{main.tex}
74
75  \clearpage
76
77  % Practical Content
78  \section{Mathematical Equations}
79
80  LaTeX is used to write mathematical expressions such as
81  fractions, powers, subscripts, quadratic equations,
82  matrices and integrals.
83
84  \subsection{Fractions}
85
86  Fractions can be written using the \verb|\frac| command.
87
88  \[
89      \frac{a+b}{c+d}
90  \]
91
92  Example:
93
94  \[
95      \frac{x+2}{x-3}=\frac{5}{2}
96  \]
97
98  \subsection{Superscripts and Subscripts}
99
100 Superscripts are created using \verb|^| and subscripts
101 are created using \verb|_|.
102
103 \[
104     x^2+y^2=z^2
105 \]
106
107 \[
108     a_1+a_2+a_3
109 \]
110
111 \[
112     x_{n+1}=x_n+h
113 \]
114
115 \subsection{Quadratic Formula}
116
117 A quadratic equation is written as:
118
119 \[
120     ax^2+bx+c=0
121 \]
122
123 Its roots are calculated using the quadratic formula:
124
125 \[
126     x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}
127 \]
128
129 \subsection{Matrices}
130
131 Matrices can be created using the \texttt{bmatrix}
132 environment. The symbol \verb|&| separates columns,
133 and \verb|\\| starts a new row.
134
135 \[
136 A=
137 \begin{bmatrix}
138     1 & 2 & 3\\
139     4 & 5 & 6\\
140     7 & 8 & 9
141 \end{bmatrix}
```

```
\]

\subsection{Integrals}

The \verb|\int| command is used to write integrals.

\[
    \int_a^b f(x)\,dx
\]

Example:

\[
    \int_0^2 x^2\,dx=\frac{8}{3}
\]

\subsection{Aligning Equations}

The \texttt{align} environment aligns multiple equations
at the ampersand symbol.

\begin{align}
    2x+3y &= 12\\
    x-y &= 1\\
    3x+2y &= 13
\end{align}

\section{Conclusion}

In this practical, we learned to write fractions,
superscripts, subscripts, the quadratic formula,
matrices and integrals. We also learned to align
multiple equations using LaTeX.

\end{document}
```

2 Mathematical Equations

LaTeX is used to write mathematical expressions such as fractions, powers, subscripts, quadratic equations, matrices and integrals.

2.1 Fractions

Fractions can be written using the `\frac` command.

$$\frac{a+b}{c+d}$$

Example:

$$\frac{x+2}{x-3} = \frac{5}{2}$$

2.2 Superscripts and Subscripts

Superscripts are created using `^` and subscripts are created using `_`.

$$x^2 + y^2 = z^2$$

$$a_1 + a_2 + a_3$$

$$x_{n+1} = x_n + h$$

2.3 Quadratic Formula

A quadratic equation is written as:

$$ax^2 + bx + c = 0$$

Its roots are calculated using the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2.4 Matrices

Matrices can be created using the `bmatrix` environment. The symbol `&` separates columns, and `\\` starts a new row.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

2.5 Integrals

The `\int` command is used to write integrals.

$$\int_a^b f(x)\,dx$$

Example:

$$\int_0^2 x^2\,dx = \frac{8}{3}$$

2.6 Aligning Equations

The `align` environment aligns multiple equations at the ampersand symbol.

$$2x + 3y = 12 \tag{1}$$
$$x - y = 1 \tag{2}$$
$$3x + 2y = 13 \tag{3}$$

3 Conclusion

In this practical, we learned to write fractions, superscripts, subscripts, the quadratic formula, matrices and integrals. We also learned to align multiple equations using LaTeX.